Layer-wise Analysis of Representation Fine-Tuning under resource constraints: Which layers matter most?

Muhammad Saad Azam
Department of Computer Science
University of Engineering and Technology
Lahore, Pakistan
saad.azam1090@gmail.com

***Abstract*—Abstract**

***Index Terms*—component, formatting, style, styling, inserts**

I. Introduction

Since the introduction of transformers [1], the landscape has changed to building large-scale language models. Today, Large Language Models (LLMs) are comprised of billions of parameters [2]. With this enormous size comes the challenging task of fine-tuning these models for specific tasks. While full finetuning still yields more accurate results, the cost of this is enormous. To address this challenge, the field of parameter-efficient finetuning (PEFT) has gathered considerable attention. These methods enable researchers to experiment on state-of-the-art models with limited computational resources.

Different techniques have been introduced to fine-tune LLMs. Adapter layers insert a small module in a frozen transformer and train it [4]. Prefix tuning modifies the inputs to attentions instead of modifying weights or activations [5]. Hu et al. [3] introduced a Low-Rank Adaptation (LoRA) of Large Language Models, a way to fine-tune a model without fine-tuning all of its parameters. All these methods either change the inputs or modify the weights without requiring full finetuning.

One more highly efficient technique of PEFT is Representation Finetuning (ReFT) for Language Models. In ReFT, the representations are directly modified instead of weights of the models [6]. Hence, ReFT acts as a drop-in intervention in representations and steer the output of the model towards a specific direction. Furthermore, ReFT trains over less parameters compared to a standard LoRA setup and achieves fairly competent performance over various tasks.

Unlike weight-based adapters which modify model's knowledge base, ReFT operates on the hidden states. ReFT helps the model learn geometric transformations which in result nudge the direction of the model's output without altering its memory. This distinction is crucial because it signals the success of these representations depends more on the semantic richness of the representations at a given depth, which can vary depending on the model architecture and the task of finetuning.

However, ReFT finetuning has a large hyperparameter search space and it is sensitive to layer selection [6]. Also, if ReFT is applied to multiple layers, the intervention matrices must be kept active during inference, introducing some inference delay. Furthermore, ReFT has not been thoroughly studied for Small Language Models (SLMs) under resource constraints on a limited dataset. The original paper also showcases how ReFT is susceptible to nature of tasks as well, further expanding the hyperparameter search space. This requires investigation on how the search space can be narrowed down based on the model architecture and the dataset constraints in place.

With this backdrop, it can be highlighted that the efficacy of ReFT is regulated by layer-wise specialization and architectural constraints. Existing work has primarily focused on 7B+ models. The behavior of ReFT in the 'Low-Resource' areas is under-explored. The research will focus on the following questions:

- What does the trend of loss looks like for the layers? Do some layers tend to overfit in ReFT while others do not?
- Does the accuracy change drastically between the layers to specific tasks?
- Are interventions productive for small language models?
- Are the accuracy and representation trends between the models similar across the layers?

II. Literature Review

A. Representation Finetuning

Representation Finetuning (ReFT) is a parameter-efficient finetuning technique for LLMs [6]. It trains interventions over the hidden states to steer model behavior into the direction of the finetuning task at inference time. Unlike other methods that target weights, ReFT modifies the hidden represenations. The main inspiration of this methodology is the study of causal interventions to direct the behavior of the model [7]. There are two types of interventions introduced originally: LoReFT and DiReFT. LoReFT learns intervention in linear subspace spanned by a low-rank projection matrix, while DiReFT simply shifts the representation by a fixed vector.

DiReFT is more straightforward and efficient but is not as accurate as LoReFT is. LoReFT transformation looks like this:

$$\Phi(h) = h + R^T(Wh + b - Rh)$$

Where h is the original hidden representation, R is the low-rank projection matrix, W is the learned weight matrix and b is the learned bias vector.

B. Linearly Decodable Features

Research has shown that many features encoded in the hidden representations are linearly decodable, simulated with linear probing experiments [8]. This is the area of interpretation that ReFT targets. When features are linearly decodable, a learned linear projection over hidden representations can be quite effective. However, this is not always the case, as other research argues the opposite [9]. This makes ReFT potentially less effective where the relevant subspace cannot be manipulated through linear interventions, varying its efficiency from task to task.

C. Model and Layer-wise Analysis

Much work has shown how layers can be specialized for certain tasks depending on the depth. Song et al., have shown how even the later layers are important for certain generation-based tasks and pruning them can degrade the performance of the LLMs within certain context and evaluation metric [10]. The original ReFT paper also identifies the large parameter space and how targetting different layers can vary the efficiency of the LLM over fine-tuning task [6]. Recent work has shown that small models exhibit a fundamentally different learning capacity under fine-tuning compared to their larger counterparts [11], suggesting that interventions calibrated for large models may not transfer directly to SLMs. This characterizes how certain layers can be more receptive to interventions towards certain tasks, especially when it comes to lighter and smaller language models.

III. Methodology

The methodology adopted in this paper follows a **controlled layer-wise intervention framework** to systematically analyze how Representation Fine-Tuning (ReFT) affects performance across different transformer depths under resource-constrained settings. For each model, ReFT is applied independently to a single transformer layer at a time, while keeping all other training variables fixed, including optimizer configuration, learning rate, batch size, dataset splits, and intervention hyperparameters. This layer-wise sweep enables isolation of the contribution of each individual layer, allowing a direct comparison of how representational interventions at different depths influence downstream task performance. The experiments are conducted on lightweight decoder-only models, namely Qwen-0.6B [12] and Granite-350M, as well as an encoder-only model, RoBERTa [14], to study whether layer-wise ReFT behavior generalizes across model families.

Evaluation is performed on a diverse set of benchmarks spanning sentiment analysis, commonsense reasoning, and natural language inference, including SST-2 [15], HellaSwag [16], MNLI [18], and QQP [17]. For decoder-only models, HellaSwag and SST-2 are evaluated using a generative single-token classification setup, where the model is prompted with the input context and corresponding answer choices, and is required to generate one token from the label space A, B, C, D; accuracy is computed based on exact match with the ground-truth label. This formulation reframes multiple-choice evaluation as constrained token generation, enabling direct assessment of how ReFT influences decision-making behavior in constrainted and regressive settings. In addition to accuracy and loss-based metrics, representational changes induced by intervention are analyzed using Centered Kernel Alignment (CKA) and Singular Value Decomposition (SVD) to study similarity structure and rank dynamics of learned intervention parameters across layers. All experiments are implemented using Google Colab and Kaggle environments, primarily utilizing NVIDIA T4 GPUs for training and evaluation.

TABLE I
HYPERPARAMETERS FOR FINE-TUNING ACROSS ALL BENCHMARK DATASETS.

Hyperparameter	**SST-2**	**HellaSwag**	**MNLI**	**QQP**
Batch Size	32	4	32	32
Learning Rate	6e-4	2.5e-4	2.5e-4	3e-4
Training Samples	1200	1200	5000	5000
Evaluation Samples	850	850	1000	1000
Optimizer	AdamW	AdamW	AdamW	AdamW
Max Length	256	512	256	128

IV. Experiments

References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 6000–6010.

[2] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, and D. Amodei, "Language models are few-shot learners," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[3] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2022.

[4] N. Houlsby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. de Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, "Parameter-efficient transfer learning for NLP," in *Proc. Int. Conf. on Machine Learning (ICML)*, 2019, pp. 2790–2799.

[5] X. L. Li and P. Liang, "Prefix-tuning: Optimizing continuous prompts for generation," in *Proc. Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021.

[6] Z. Wu, A. Arora, Z. Wang, A. Geiger, D. Jurafsky, C. D. Manning, and C. Potts, "ReFT: Representation Finetuning for Language Models," *arXiv preprint arXiv:2404.03592*, 2024.

[7] A. Geiger, Z. Wu, C. Potts, T. Icard, and N. D. Goodman, "Finding Alignments Between Interpretable Causal Variables and Distributed Neural Representations," *arXiv preprint arXiv:2303.02536*, 2023.

[8] W. Gurnee and M. Tegmark, "Language Models Represent Space and Time," *arXiv preprint arXiv:2310.02207*, 2023.

[9] J. Engels, E. J. Michaud, I. Liao, W. Gurnee, and M. Tegmark, "Not All Language Model Features Are Linear," *arXiv preprint arXiv:2405.14860*, 2024.

[10] X. Song, K. Wang, P. Li, L. Yin, and S. Liu, "Demystifying the Roles of LLM Layers in Retrieval, Knowledge, and Reasoning," *arXiv preprint arXiv:2510.02091*, 2025.

[11] Y. Li, X. Yue, Z. Xu, F. Jiang, L. Niu, B. Y. Lin, B. Ramasubramanian, and R. Poovendran, "Small Models Struggle to Learn from Strong Reasoners," *arXiv preprint arXiv:2502.12143*, 2025.

[12] A. Yang, A. Li, B. Yang, et al., "Qwen3 technical report," *arXiv preprint arXiv:2505.09388*, 2025.

[13] Gemma Team, "Gemma 3 technical report," *arXiv preprint arXiv:2503.19786*, 2025.

[14] Y. Liu, M. Ott, N. Goyal, et al., "RoBERTa: A robustly optimized BERT pretraining approach," *arXiv preprint arXiv:1907.11692*, 2019.

[15] R. Socher, A. Perelygin, J. Wu, J. Chuang, C. D. Manning, A. Ng, and C. Potts, "Recursive deep models for semantic compositionality over a sentiment treebank," in *Proc. Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, 2013.

[16] R. Zellers, A. Holtzman, Y. Bisk, A. Farhadi, and Y. Choi, "HellaSwag: Can a machine really finish your sentence?" *arXiv preprint arXiv:1905.07830*, 2019.

[17] S. Iyer, N. Dandekar, and K. Csernai, "First Quora dataset release: Question pairs," Quora, 2017.

[18] A. Williams, N. Nangia, and S. Bowman, "A broad-coverage challenge corpus for sentence understanding through inference," *arXiv preprint arXiv:1704.05426*, 2017.